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Last Minute Notes for Computer Network

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A computer network is a system of interconnected computing devices that communicate with one another over wired or wireless communication links to exchange data, share resources, and deliver services, using a well-defined set of communication protocols.

Computer networks form the foundation of modern communication systems, including the Internet, cloud services, and distributed applications.

Objectives of Computer Networks

The main goals of computer networking are:

- **Resource Sharing** – sharing hardware (printers, storage), software, and data
- **Communication** – email, messaging, video conferencing
- **Reliability** – backup paths and fault tolerance
- **Scalability** – easy expansion as the number of users grows
- **Cost Efficiency** – reduced cost through shared infrastructure

Basics of Computer Networks

Network Components

Every computer network consists of the following fundamental components:

- **Sender (Source)** – device that generates data
- **Receiver (Destination)** – device that receives data
- **Message** – information being transmitted
- **Transmission Medium** – physical or wireless path for data transfer
- **Protocol** – set of rules governing communication

Network Topologies

Network topologies refer to the physical or logical arrangement of devices (nodes) and communication links in a computer network. They define how devices are connected to each other and how data flows within the network. There are different topologies which are used in computer network:

Mesh Topology

- Every device is directly connected to every other device using a dedicated communication link.
- For N devices, the total number of links required is: Number of links=(N(N−1))/2
- Provides high reliability and fault tolerance due to multiple alternative paths.
- Expensive and complex to install and maintain because of a large number of links.

Bus Topology

- All devices are connected to a single shared communication channel called the backbone cable.
- For N devices one main cable is required and it is connected by drop lines with each individual device.
- Data transmitted by a device is broadcast to all devices on the network.
- Simple and cost-effective to implement.
- Less reliable and inefficient for large networks.

Star Topology

- All devices are connected to a central device (hub or switch) using point-to-point cables.
- For N devices, N cables are required.
- Easy to install, manage, and troubleshoot.
- Failure of a single device does not affect the rest of the network.
- Failure of the central device can cause the entire network to stop working.

Ring Topology

- Each device is connected to exactly two neighboring devices, forming a closed loop.
- Data travels in a circular direction (clockwise or counterclockwise).
- Provides orderly and controlled data transmission.
- Failure of a single device or link can disrupt the entire network

Read more about Network Topologies.

Transmission Modes

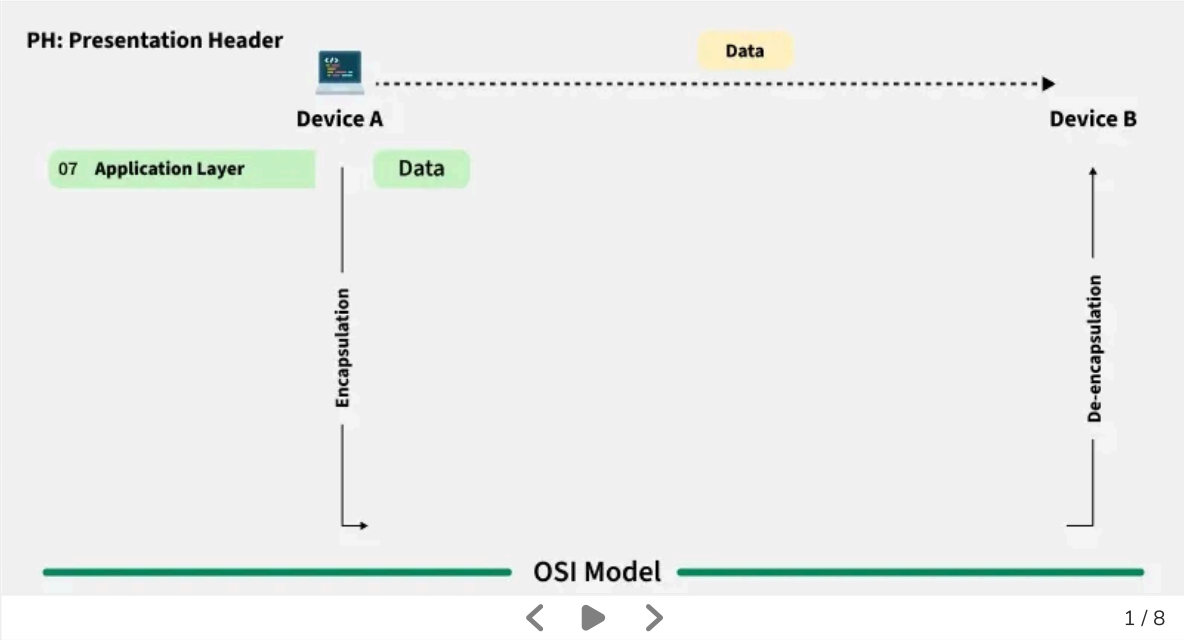
- **Simplex Mode** : the communication is unidirectional, as on a one-way street. Only one of the two devices on a link can transmit, and the other can only receive.
- **Half-duplex Mode** : each station can both transmit and receive, but not at the same time.

- **Full-duplex Mode** : both stations can transmit and receive simultaneously.

Read more about Transmission modes.

OSI Model

OSI stands for **Open Systems Interconnection**. It has been developed by ISO– **International Organization for Standardization**, in the year 1984. It is a seven-layer architecture with each layer having specific functionality to perform.



Understanding Layers of OSI Model

Physical Layer

The Physical Layer is the lowest layer of the OSI model and is responsible for the transmission of raw bits (0s and 1s) over a physical communication medium.
It defines the mechanical, electrical, and procedural characteristics required to establish and maintain physical connections between devices.

Key Responsibilities of the Physical Layer

- Bit transmission over the physical medium
- Defines voltage levels for binary data
- Specifies data rate (bits per second)
- Determines bit synchronization
- Defines line configuration (point-to-point or multipoint)
- Specifies physical topology
- Defines transmission modes (simplex, half duplex, full duplex)

Transmission Media

Transmission media refer to the physical or wireless paths through which data travels from the sender to the receiver in a computer network. They are mainly classified into guided (wired) and unguided (wireless) media.

Guided Transmission Media (Wired)

Guided media use a physical path (cables) to transmit data signals.

Characteristics

- Data travels through solid conductors
- More secure and reliable
- Less interference and noise
- Limited mobility

Types

- **Twisted Pair Cable** – low cost, commonly used in LANs
- **Coaxial Cable** – better noise resistance, used in cable TV
- **Optical Fiber** – very high speed and bandwidth, long-distance communication

Non-Guided Transmission Media (Wireless)

Non-guided media transmit data using electromagnetic waves without physical cables.

Characteristics

- No physical connection required
- Supports mobility and flexibility
- Easier and faster deployment
- More prone to interference and security risks

Types

- **Radio Waves** – Wi-Fi, Bluetooth
- **Microwaves** – cellular and satellite communication
- **Infrared** – short-range communication

Switching Techniques

Switching techniques are methods used in a network to establish a path and transfer data from a source to a destination through intermediate nodes.

Circuit Switching

- In circuit switching, dedicated communication path is established for the entire communication session.
- Connection setup → data transfer → connection termination
- Fixed bandwidth throughout the communication
- Example: Traditional telephone networks (PSTN).
- Advantages: Predictable and continuous data flow.
- Disadvantages: Inefficient use of resources, path reserved even when idle.

Message Switching

- In message switching, entire message is sent to an intermediate node, stored, and forwarded until it reaches the destination.
- No dedicated path required
- Example: Email systems.
- Advantages: Efficient use of resources and no dedicated path.
- Disadvantages: Delays due to storing and forwarding large messages.

Packet Switching

- In packet switching, data is divided into packets that are routed independently to the destination.
- No dedicated path required
- Packets may follow different routes
- Example: The Internet (TCP/IP).
- Advantages: Efficient utilization of bandwidth, flexible, scalable.
- Disadvantages: Potential for packet reordering, overhead for packet headers.

Comparison of Switching Techniques

Circuit Switching	Packet Switching	Message Switching
Uses a dedicated end-to-end path	No dedicated path is established	No dedicated path is used
Data is transmitted as a continuous stream	Data is divided into packets	Data is sent as a complete message
Delay is low after connection setup	Delay is variable due to routing and queuing	High delay due to store-and-forward
Bandwidth utilization is low	Bandwidth utilization is high	Bandwidth utilization is low
Used in traditional telephone networks	Used in the Internet	Used in old telegraph systems

Data Link Layer

It is the second layer of the OSI model and is responsible for node-to-node delivery of data. It ensures reliable transmission of frames over a physical link by handling framing, error control, flow control, and medium access control.

Functions of Data Link Layer

- **Framing** – divides raw bit stream into frames
- **Error control** – detects and corrects transmission errors
- **Flow control** – prevents sender from overwhelming receiver
- **Medium Access Control (MAC)** – controls access to shared medium
- **Physical addressing** – uses MAC addresses

Basic Time Delays

1. Transmission Delay (Td)

Time taken to place all bits of a frame onto the transmission medium.

Td = Length of packet / bandwidth

2. Propagation Delay (Pd)

Time taken by a signal to travel from sender to receiver.

Pd = Distance / Propagation speed

3. Round Trip Time (RTT)

- Total time taken for a frame to reach the receiver and for its acknowledgment to return

- Also known as minimum acknowledgment waiting time.

Flow Control Mechanisms

Flow control ensures that the sender does not transmit data faster than the receiver can handle.

1. Stop and Wait Protocol

- Only one frame is sent at a time
- Sender waits for ACK before sending the next frame
- Simple but results in poor utilization

Efficiency (Line Utilization):

E = Td(frame) / Td(frame)+2Pd+Qd+Prd+Td(ACK)

Throughput:

Throughput =L / Td(frame)+2Pd+Qd+Prd+Td(ACK)

2. Go-Back-N ARQ

- Also called a conservative protocol
- Uses cumulative acknowledgments
- Receiver window size = 1
- Sender window size = $2^k - 1$, where *k* is number of bits in sequence number

Efficiency:

E = N×Td(frame) / Td(frame)+2Pd+Qd+Prd+Td(ACK)

Throughput:

Throughput = N×L / Td(frame)+2Pd+Qd+Prd+Td(ACK)

3. Selective Repeat ARQ

- Retransmits only lost or corrupted frames
- More efficient than Go-Back-N
- Reduces unnecessary retransmissions
- Higher buffer requirement

Efficiency:

E = N×Td(frame) / Td(frame)+2Pd+Qd+Prd+Td(ACK)

Properties	Stop and wait	Go back N	Selective repeat
efficiency	1 / (1+2a)	N / (1+2a)	N / (1+2a)
buffer	1 + 1	N + 1	N + N
sequence number	1 + 1	N + 1	N + N
retransmission	1	N	1
bandwidth	low	High	Moderate
CPU	low	Moderate	High
implementation	low	Moderate	Complex

Comparison of Data Link Layer Protocols

Framing in DLL : Framing provides a way for a sender to transmit a set of bits that are meaningful to the receiver.

- **Character/Byte Stuffing:** Used when frames consist of character. If data contains ED then, a byte is stuffed into the data to differentiate it from ED.
- **Bit stuffing** : Sender stuffs a bit to break the pattern i.e. here appends a 0 in data = 0111 **0** 1.

Medium Access Control Protocols

They manage how devices share a common communication medium and ensure efficient and collision-free data transmission. MAC protocols are generally classified into the following categories:

Contention-Based Protocols

In these protocols, devices compete for the medium, and collisions may occur. They are suitable for networks with more traffic.

Examples :

ALOHA

- **In** Pure ALOHA, devices transmit data whenever they want. Collisions are resolved using retransmissions.
- **In** Slotted ALOHA, the transmission time is divided into slots, reducing the probability of collisions.

CSMA (Carrier Sense Multiple Access)

- **In** CSMA, devices sense the medium before transmitting.
- **CSMA/CD (Collision Detection)** is used in Ethernet, it detects collisions and retransmits data.
- **CSMA/CA (Collision Avoidance)** is used in wireless networks (e.g., Wi-Fi), it avoids collisions by sending RTS/CTS (Request to Send/Clear to Send) signals.

Controlled Access Protocols

These protocols avoid collisions by controlling access to the medium. Examples :

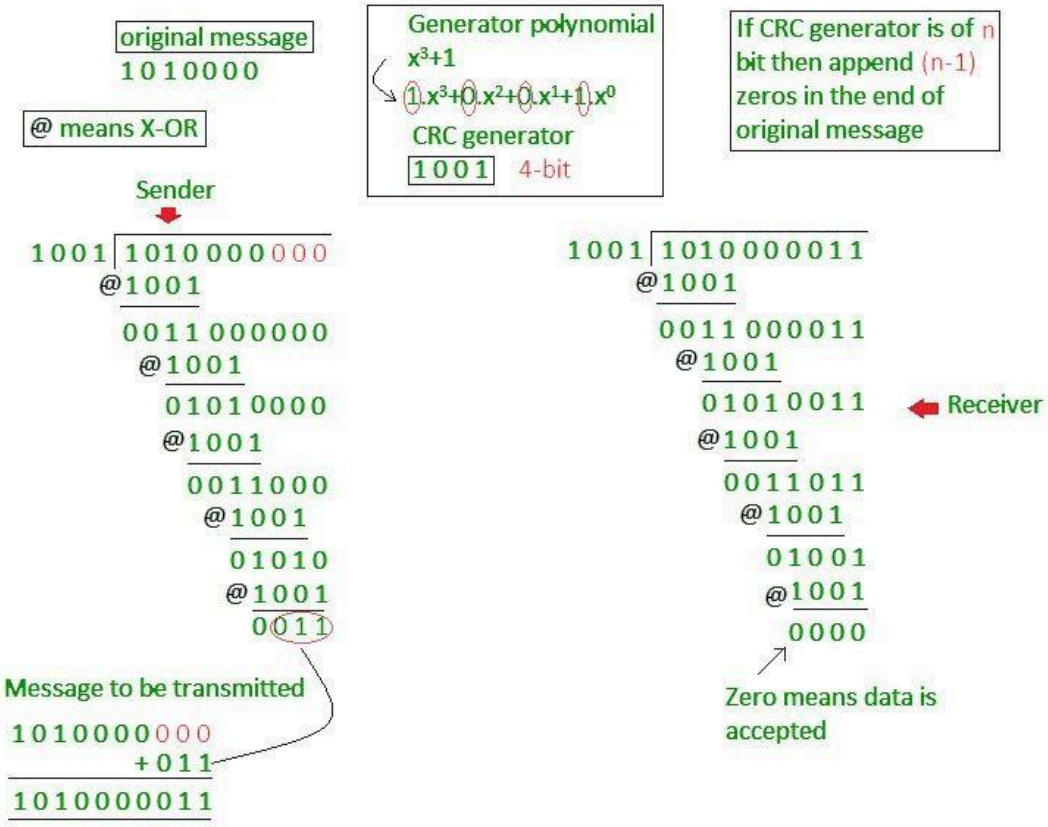
- In polling, a central device polls others to grant them access to the medium.
- **Token Passing**: A token circulates in the network, and only the device holding the token can transmit.

Error Control

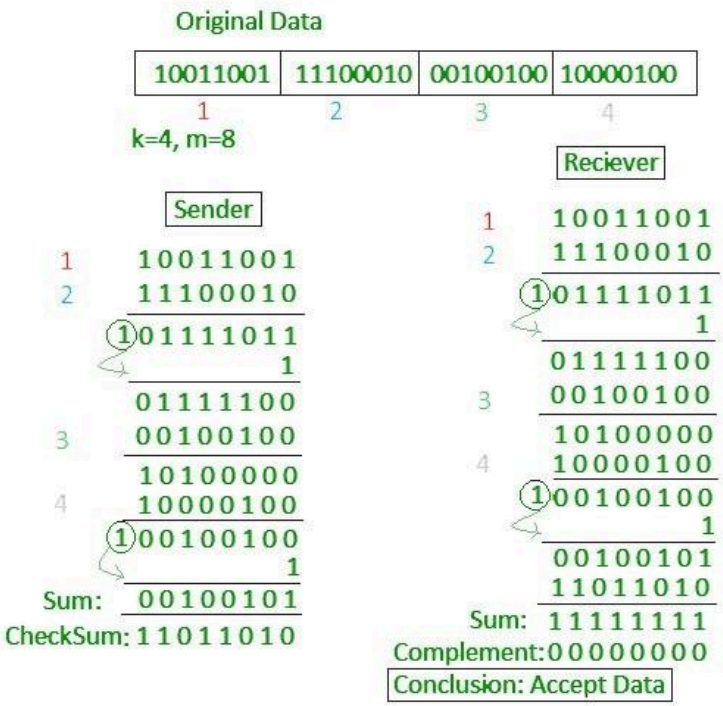
- **Hamming Code**: Hamming Code is a set of error-correction codes that can be used to detect and correct the errors that can occur when the data is moved or stored from the sender to the receiver.

```
Redundant bits: 2  r  ≥ m + r + 1
where, r = redundant bit, m = data bit
```

- **Cyclic Redundancy Check** : CRC uses a polynomial generator on both the sender and receiver sides to detect the error in the transmitted data. Example: Let's data to be send is 1010000 and divisor in the form of polynomial is x^3+1. CRC method discussed below.



- **Checksum**: Checksum is more reliable than other modes of error detection, it uses a checksum generator on the sender side and a checksum checker on the receiver side. Example:



Network Layer

The Network Layer is responsible for logical addressing, routing, and packet forwarding across different networks. It ensures that data reaches the correct destination, even across multiple routers.

Class Full Addressing is an IP addressing method that divides the address space into five classes: **A, B, C, D, and E**, based on the first few bits of the IP address. Each class has a fixed network and host portion.

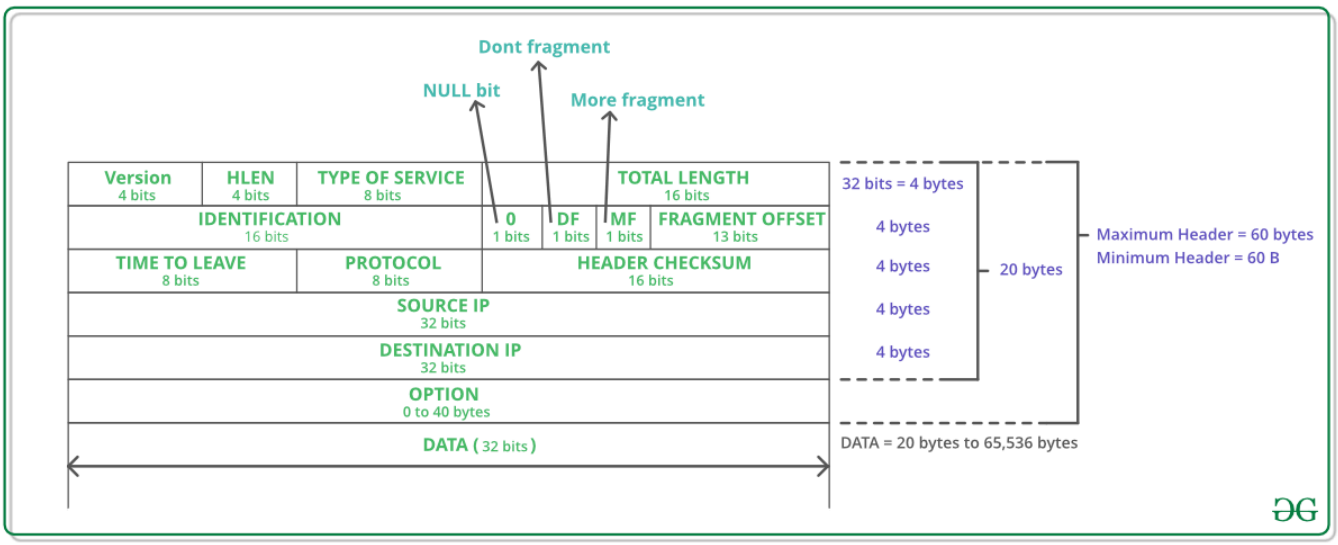
Address Class	First octet (Decimal)	First octet bits (red bits don't change)	Network (N) and Host (H) portion	Default Subnet Mask
A	1 – 127	0000 0000 – 0111 1111	N.H.H.H	255.0.0.0
B	128 – 191	1000 0000 – 1011 1111	N.N.H.H	255.255.0.0
C	192 – 223	1100 0000 – 1101 1111	N.N.N.H	255.255.255.0
D	224 – 239	1110 0000 – 1110 1111	n/a (multicast)	
E	240 – 255	1111 1111 – 1111 1111	n/a (experimental)	

Class Full Addressing

IPv4 Datagram Header

The IPv4 header contains information required to **route, fragment, and control packets** across networks.

- **VERSION:** Version of the IP protocol (4 bits), which is 4 for IPv4
- **HLEN:** IP header length (4 bits), which is the number of 32 bit words in the header. The minimum value for this field is 5 and the maximum is 15.
- **Type of service:** Low Delay, High Throughput, Reliability (8 bits)
- **Total Length:** Length of header + Data (16 bits), which has a minimum value 20 bytes and the maximum is 65,535 bytes.
- **Identification:** Unique Packet Id for identifying the group of fragments of a single IP datagram (16 bits)
- **Flags:** 3 flags of 1 bit each : reserved bit (must be zero), do not fragment flag, more fragments flag (same order)
- **Fragment Offset:** Represents the number of Data Bytes ahead of the particular fragment in the particular Datagram. Specified in terms of number of 8 bytes, which has the maximum value of 65,528 bytes.
- **Time to live:** Datagram's lifetime (8 bits), It prevents the datagram to loop through the network by restricting the number of Hops taken by a Packet before delivering to the Destination.
- **Protocol:** Name of the protocol to which the data is to be passed (8 bits)
- **Header Checksum:** 16 bits header checksum for checking errors in the datagram header
- **Source IP address:** 32 bits IP address of the sender
- **Destination IP address:** 32 bits IP address of the receiver
- **Option:** Optional information such as source route, record route. Used by the Network administrator to check whether a path is working or not.



Subnetting

Subnetting is the process of dividing a large IP network into smaller logical networks (subnets). It improves network management, security, performance, and efficient IP address utilization.

Why Subnetting is Needed

- Reduces broadcast traffic
- Improves network security
- Efficient utilization of IP addresses
- Simplifies routing and network management

Implementation :

Let's consider a network **192.168.1.0/24** (subnet mask: 255.255.255.0), we need to create **4 subnets**.

- **Find the Number of Subnets**:

```
2^n ≥ Number of Subnets
Here, n=2 (since 2^2 = 4).
```

- **Determine the New Subnet Mask**:

A subnet mask is a 32-bit number used in IPv4 to divide an IP address into:

Network portion

Host portion

Routers use the subnet mask to identify the network address and determine routing decisions.

```
Original subnet mask: /24
Add n=2 bits for subnetting: /26
New subnet mask: 255.255.255.192
```

- **Find Subnet Ranges**:

```
Subnet size = 2^32−New Prefix=2^(32−26)=64 IP addresses per subnet.
Usable IPs per subnet = 64−2=62 (excluding network and broadcast addresses).
```

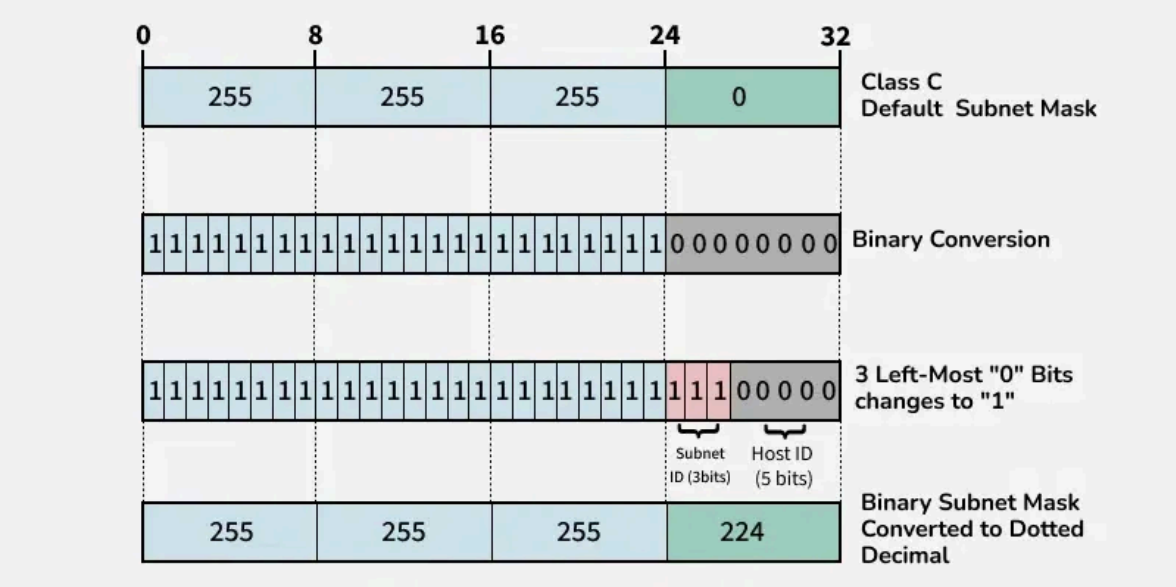
- **Subnets**:

```
Subnet 1: 192.168.1.0 to 192.168.1.63
Subnet 2: 192.168.1.64 to 192.168.1.127
Subnet 3: 192.168.1.128 to 192.168.1.191
Subnet 4: 192.168.1.192 to 192.168.1.255
```

Each subnet has:

- **64 total addresses**
- **62 usable IPs**
- Subnet mask: **255.255.255.192 (/26)**

A **subnet mask** is a 32-bit number used in IPv4 to divide an IP address into network and host portions. It helps identify the size of the network and enables subnetting by determining which part of the address belongs to the network and which to the host.



Subnet Mask of Class C IP address

Supernetting

Supernetting is the process of combining multiple smaller networks into a single larger network.

Purpose

- Reduces the size of routing tables
- Enables route summarization
- Improves routing efficiency and scalability

Key Points

- Opposite of subnetting
- Uses a shorter prefix length
- Commonly used by routers and ISPs

VLSM

VLSM: It allows the use of different subnet masks within the same network, meaning subnets can have different sizes based on requirements.

Characteristics

- Multiple subnet masks in one network
- Subnetting of an already subnetted network
- Supports variable-sized subnets

Advantages

- Efficient use of IP addresses
- Reduces IP wastage
- Flexible network design

Internet Control Message Protocol:

The Internet Protocol (IP) does not provide error reporting or control messages.

ICMP is used to report errors and provide network diagnostics.

Functions

- Reports delivery errors
- Handles network congestion notifications
- Supports diagnostic tools

Common ICMP Messages

- Destination Unreachable
- Time Exceeded
- Echo Request / Echo Reply (Ping)

Difference between DVR and LSR:

Distance Vector Routing (DVR)	Link State Routing (LSR)
Routers share routing information with neighbors only	Routers share link-state information with all routers
Based on local knowledge	Based on global network knowledge
Less bandwidth (small updates, no flooding)	More bandwidth (flooding of link-state packets)
Periodic updates	Event-based updates
Bellman–Ford Algorithm	Dijkstra's Algorithm
Slow	Fast
Count-to-infinity problem exists	No count-to-infinity problem
Persistent routing loops possible	Only temporary (transient) loops
Not suitable for large networks	Suitable for large networks
RIP, IGRP	OSPF, IS-IS

Hop Count (Routing Metric)

Hop count is the number of routers a packet must pass through to reach the destination network.

- Path with the lowest hop count is considered the best route
- Used as a routing metric
- Simple but not always optimal for large networks

Used by: Distance Vector Routing protocols such as RIP

Open Shortest Path First

OSPF is a link-state routing protocol that finds the best path between source and destination routers using the Shortest Path First (SPF) algorithm.

Key Characteristics

- Uses Dijkstra's algorithm
- Maintains a complete network topology
- Metric based on cost (bandwidth)
- Fast convergence and scalable

Designated Router(DR) and Backup Designated Router(BDR) election takes place in the broadcast network or multi-access network.

Routing Information Protocol

RIP is a distance vector routing protocol that uses hop count as its routing metric.

Key Characteristics

- Maximum hop count: 15
- Administrative Distance: 120
- Uses UDP port 520
- Works at the application layer of the OSI model
- Simple but not suitable for large networks

Routing Information Protocol (RIP): RIP is a dynamic routing protocol that uses hop count as a routing metric to find the best path between the source and the destination network. It is a distance vector routing protocol that has an AD value of 120 and works on the application layer of the OSI model. RIP uses port number 520.

ARP (Address Resolution Protocol)

An **ARP (Address Resolution Protocol)**request is a network protocol used to map an IP address to its corresponding MAC (Media Access Control) address within a local network.

RARP Protocol (Reverse Address Resolution Protocol)

Reverse Address Resolution Protocol (RARP) is a network protocol used to obtain an IP address for a device (like a diskless workstation) from its MAC address

Transport Layer

- The Transport Layer is responsible for end-to-end communication, process-to-process delivery, reliability, flow control, and congestion control.

Transmission Control Protocol (TCP)

- TCP is a connection-oriented, reliable transport protocol that ensures ordered and error-free data delivery.
- **TCP Header :** The TCP header is a fixed 20-byte (minimum) structure in the Transport Layer that includes fields such as source and destination ports, sequence and acknowledgment numbers, flags (e.g., SYN, ACK), and window size to ensure reliable and ordered data delivery.

TCP Congestion Control

TCP Congestion Control manages network congestion by adjusting the data transmission rate **.**It ensures efficient and reliable data delivery while avoiding network overload. It uses algorithms like:

- Slow Start Phase : In this phase after every RTT the congestion window size increments exponentially
- Congestion Avoidance Phase : This phase starts after the threshold value also denoted as ssthresh. The size of CWND (Congestion Window) increases additive. After each RTT cwnd = cwnd + 1.
- Congestion Detection Phase : If congestion occurs, the congestion window size is decreased.

TCP 3-Way Handshake Process

The TCP 3-Way Handshake Process is implemented in following steps:

- **Step 1 (SYN):** The client wants to establish a connection with the server, so it sends a segment with SYN(Synchronize Sequence Number) which informs the server that the client is likely to start communication and with what sequence number it starts segments with

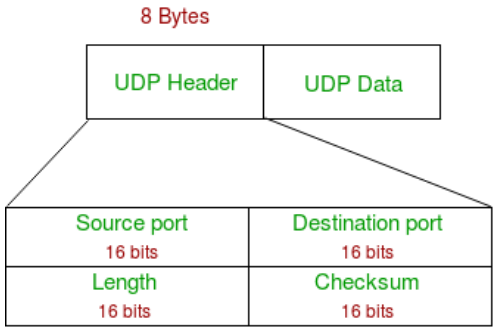
- **Step 2 (SYN + ACK)** : Server responds to the client request with SYN-ACK signal bits set. Acknowledgment (ACK) signifies the response of the segment it received and SYN signifies with what sequence number it is likely to start the segments with
- **Step 3 (ACK):** In the final part client acknowledges the response of the server and they both establish a reliable connection with which they will start eh actual data transfer.

TCP Connection Termination

- TCP connection termination is a process that ends a Transmission Control Protocol (TCP) connection. TCP supports two types of connection releases like most connection-oriented transport protocols:
- **Graceful connection release: T**he connection is open until both parties have closed their sides of the connection.
- **Abrupt connection release: E**ither one TCP entity is forced to close the connection or one user closes both directions of data transfer.

User Datagram Protocol (UDP)

- UDP is a connectionless transport protocol that provides fast data transmission with minimal overhead. It does not guarantee reliability, ordering, or flow control.
- **UDP Header :**The UDP header is a simple 8-byte structure in the Transport Layer, containing four fields: Source Port, Destination Port, Length, and Checksum



Difference between TCP and UDP

TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Connection-oriented protocol	Connectionless protocol
Reliable data delivery using acknowledgments and retransmissions	Unreliable data delivery (no acknowledgments)
Ensures in-order delivery using sequence numbers	No guarantee of packet order
Uses checksum, ACKs, and retransmission	Uses checksum only (optional in IPv4)
Supported using sliding window mechanism	Not supported
Implemented using Slow Start, Congestion Avoidance, Fast Retransmit	Not supported

TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Slower due to overhead	Faster due to minimal overhead
Minimum 20 bytes	Fixed 8 bytes
Uses 3-way handshake (SYN, SYN-ACK, ACK)	No handshake
Byte-stream oriented	Message-oriented
Detects and retransmits lost packets	Lost packets are not recovered
High overhead due to control mechanisms	Low overhead
HTTP, HTTPS, FTP, SMTP, SSH	DNS, VoIP, Video Streaming, Online Games

Session Layer

The Session Layer is the 5th layer of the OSI model. Its primary role is to establish, manage, and terminate sessions between two communicating devices.
A session refers to a logical connection that exists for the duration of communication.

Why the Session Layer Is Needed

While the Transport Layer ensures reliable data transfer, the Session Layer manages the dialog itself:

- Who starts communication
- How long communication lasts
- When to pause, resume, or terminate communication

Key Functions of the Session Layer

Session Establishment

- Initializes a session between two network entities.
- Defines session parameters such as: Authentication, Authorization & Session ID
- Example: Logging into a remote server.

Session Maintenance

- Keeps the session active during data exchange.
- Ensures that communication continues smoothly for the entire session duration.
- Manages session checkpoints and controls data flow.

Session Termination

- Gracefully closes the session once communication is complete.
- Frees resources allocated to the session.
- Prevents unauthorized reuse of session data.

Synchronization

- Adds synchronization points (checkpoints) into the data stream.
- If a failure occurs, communication can resume from the last checkpoint, not from the beginning.

- Very useful in long data transfers.

Session Identification

- Identifies entities participating in a session.
- Uses session identifiers to track ongoing communication.
- Helps differentiate between multiple sessions from the same user or device.

Presentation Layer

The Presentation Layer is the 6th layer of the OSI model. Its primary role is to prepare data for the Application Layer by ensuring that information sent from one system can be understood by another system, regardless of differences in data format, encoding, or security mechanisms.

In simple terms, this layer acts as a translator and formatter for network data.

Why the Presentation Layer Is Needed

Different systems may use:

- Different data formats
- Different character encoding schemes
- Different encryption standards

The Presentation Layer ensures compatibility and interoperability between these systems.

Key Functions of the Presentation Layer

Data Translation (Format Conversion)

- Converts data from the sender's format into a common network format.
- At the receiver side, it converts data back into the receiver's format.
- Enables communication between heterogeneous systems.

Examples

- ASCII ↔ Unicode
- EBCDIC ↔ ASCII
- Little-endian ↔ Big-endian

Character Encoding

- Defines how characters are represented in binary form.
- Ensures correct interpretation of text data.

Common encoding schemes

- ASCII
- Unicode
- UTF-8
- UTF-16

Data Encryption and Decryption

- Protects data from unauthorized access.
- **Encryption** at the sender side converts plaintext into ciphertext.
- **Decryption** at the receiver side converts ciphertext back into readable data.

Used for

- Data confidentiality
- Secure communication (e.g., SSL/TLS)

Data Compression and Decompression

- Reduces the size of data before transmission.
- Improves bandwidth utilization and transmission speed.
- Decompression restores the original data at the receiver.

Types

- Lossless compression (e.g., ZIP, PNG)
- Lossy compression (e.g., JPEG, MP3)

Application Layer

The Application Layer is the topmost layer of the OSI model and provides network services directly to end-user applications. It enables users and software applications to access network resources, exchange data, and perform communication tasks such as email, file transfer, and web browsing.

> *The Application Layer does not include the user interface itself, but it provides services used by applications like browsers, email clients, and FTP clients.*

Functions of Application Layer

Network Resource Access : It Allows users to access remote resources such as

- Files
- Printers
- Web servers

Supports services like file sharing and directory access.

Email Services

- Enables sending, receiving, and managing emails.
- Supports protocols for mail transfer, retrieval and synchronization.

File Transfer and Management

- Supports uploading, downloading, and managing files on remote systems.
- Ensures controlled and reliable file exchange.

Name Resolution

- Translates human-readable names into machine-readable addresses.
- Makes the internet user-friendly.

Application-Level Communication

- Enables communication between client and server applications.
- Supports request–response mechanisms.

TCP/IP Model

The TCP/IP model is a practical and implementation-oriented networking model used in real-world networks, including the Internet. It was developed by DARPA (Defense Advanced Research Projects

Agency) and forms the foundation of modern data communication.

Unlike the OSI model (7 layers), the TCP/IP model consists of 4 layers, each responsible for a specific set of networking functions.

Layers of TCP/IP Model

1. Application Layer
2. Transport Layer
3. Internet Layer
4. Network Access Layer

Application Layer in TCP/IP Model

The Application Layer is the topmost layer of the TCP/IP model. It provides network services directly to user applications and enables communication between software applications over a network.

In the TCP/IP model, this layer combines the responsibilities of three OSI layers:

- Application Layer (OSI Layer 7)
- Presentation Layer (OSI Layer 6)
- Session Layer (OSI Layer 5)

Key Functions

User-Level Network Services

- Enables applications to access network resources
- Supports web browsing, email, file transfer, and remote login

Data Representation and Formatting: It handles

- Data translation
- Character encoding
- Data compression
- Encryption and decryption

Session Management

- Establishes, maintains, and terminates communication sessions
- Supports synchronization and dialog control

Name Resolution

- Converts domain names into IP addresses
- Makes network usage human-friendly

Application Communication

- Enables client–server interaction
- Supports request–response and message-based communication

Features of Application Layer (TCP/IP)

- Closest layer to the user
- Protocol-oriented

- Platform-independent
- Supports multiple application services
- Does not handle packet delivery or routing

Transport Layer in TCP/IP Model

The Transport Layer is the second layer of the TCP/IP model. It is responsible for end-to-end (process-to-process) communication between applications running on different hosts.

It ensures that data sent from a source application is delivered to the correct destination application reliably or efficiently, depending on the protocol used.

Key Functions of the Transport Layer

Process-to-Process Communication

- Uses port numbers to identify source and destination applications.
- Enables multiple applications to communicate simultaneously on the same host.

Segmentation and Reassembly

- Breaks large data into smaller units called segments.
- Reassembles segments at the receiver side.

Multiplexing and Demultiplexing

- **Multiplexing**: Multiple applications send data using a single network connection.
- **Demultiplexing**: Data is delivered to the correct application using port numbers.

Flow Control

- Ensures the sender does not exceed the receiver's processing capacity.
- Uses sliding window mechanism (in TCP).

Error Control

- Detects errors using checksums.
- Retransmits lost or corrupted data (TCP).

Congestion Control

- Adjusts data transmission rate based on network congestion.
- Prevents network overload.

Internet Layer in TCP/IP Model

The Internet Layer is the third layer of the TCP/IP model. It is responsible for logical addressing, packet routing, and delivery of data across interconnected networks.

This layer determines how packets move from the source host to the destination host, even if both are on different networks.

Key Functions

Logical Addressing

- Assigns IP addresses to devices.

- Ensures unique identification of hosts across networks.
- Supports IPv4 (32-bit) and IPv6 (128-bit) addressing.

Packet Routing

- Determines the best path from source to destination.
- Uses routing algorithms and routing tables in routers.

Packet Forwarding

- Moves packets from one router to the next until they reach the destination.
- Uses next-hop routing.

Fragmentation and Reassembly

- Breaks large packets into smaller fragments to match MTU size.
- Reassembles fragments at the destination host.

Error Reporting and Diagnostics

Reports network-level errors such as:

- Destination unreachable
- Time exceeded
- Helps in network troubleshooting.

Network Access Layer in TCP/IP Model

The Network Access Layer is the lowest layer of the TCP/IP model. It is responsible for physical transmission of data over the network medium and defines how data is placed onto and received from the physical network.

This layer combines the functionalities of the Data Link Layer and Physical Layer of the OSI model.

Key Functions

Framing

Encapsulates IP packets into **frames**

Adds:

- Source MAC address
- Destination MAC address
- Error detection information

Physical Addressing (MAC Addressing)

- Uses MAC addresses (48-bit) for local delivery.
- Ensures correct delivery within a LAN.

Media Access Control

- Determines how devices share a common transmission medium.
- Prevents or manages collisions.

Examples:

- CSMA/CD (Ethernet)
- CSMA/CA (Wi-Fi)

Error Detection

Detects transmission errors using mechanisms like:

- CRC (Cyclic Redundancy Check)

Error correction is **not** performed at this layer.

Bit Transmission

Converts frames into binary signals (bits).

Defines:

- Voltage levels
- Data rate
- Transmission mode

Difference Between OSI Model and TCP/IP Model

OSI Model	TCP/IP Model
Open Systems Interconnection	Transmission Control Protocol / Internet Protocol
ISO (International Organization for Standardization)	DARPA (U.S. Department of Defense)
7 layers	4 layers
Conceptual and theoretical model	Practical and implementation-based model
Strict layered architecture	Flexible layered architecture
Protocol-independent	Protocol-oriented
Supports only connection-oriented service	Supports both TCP (reliable) and UDP (unreliable)
Supports both connection-oriented and connectionless services	Supports only connectionless service (IP)
Implemented at Transport Layer	Implemented mainly by TCP
Separate layers	Combined into Application Layer
Data Link and Transport layers	Transport layer (TCP)
Defined in Network Layer	Defined in Internet Layer
Educational and reference model	Used in real-world networks (Internet)
Rarely implemented fully	Widely implemented (Internet)

Common Network Protocols: Ports, Layers, and Functions

Protocol (Port Number)	OSI / TCP-IP Layer	Work / Function
HTTP (80)	Application	Transfers web pages using request–response model
HTTPS (443)	Application	Secure web communication using encryption (TLS/SSL)
FTP (21 control, 20 data)	Application	Transfers files between client and server
SMTP (25)	Application	Sends emails between mail servers
IMAP (143)	Application	Accesses and synchronizes emails on server
DNS (53 UDP/TCP)	Application	Resolves domain names to IP addresses
Telnet (23)	Application	Remote login (unencrypted)
SSH (22)	Application	Secure remote login and command execution
SNMP (161/162)	Application	Network monitoring and management
DHCP (67 server, 68 client)	Application	Automatically assigns IP configuration
TCP	Transport	Reliable, ordered, connection-oriented data delivery
UDP	Transport	Fast, connectionless data delivery
IP	Internet / Network	Logical addressing and packet routing
ICMP	Internet / Network	Error reporting and diagnostics
ARP	Internet / Network	Maps IP address to MAC address
Ethernet	Data Link / Network Access	Framing and MAC-based delivery
Wi-Fi / 802.11	Data Link / Network Access	Wireless LAN communication

Network Devices

Network Devices are hardware components used to connect, manage, control, and communicate data between computers and other devices within a network. They enable data transmission, routing, security, and network management.

See Last Minute Notes on all subjects here.

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